

Ajay Srinivasan | Curriculum Vitae

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Education

University of Michigan

PhD Mathematics, CGPA: 4.0

Ann Arbor, MI

2025 – Present

University of Southern California

BS Mathematics (Honors Program), CGPA: 3.95

Los Angeles, CA

2021 – 2025

Minor in Physics

Graduate Coursework at USC: Algebra I-II, Algebraic Topology, Differential Geometry, Complex Analysis, Thermodynamics and Statistical Mechanics, Quantum Field Theory II (at Caltech), Topics in Algebraic Geometry (audited, instructor: Joseph Helfer), Seminar in Algebra: Derived ∞ -Categories (audited, instructor: Aravind Asok).

Experience

Academic

Department of Mathematics, The University of Chicago

Participant, Mathematics REU 2024

Chicago, IL

Summer 2024

Worked on operads and infinite loop spaces in motivic homotopy theory

Dept. of Physics and Astronomy, University of Southern California

Undergraduate Researcher

Los Angeles, CA

2022–2024

Worked on the theory and computational simulation of binary Bose-Einstein condensates in two dimensions.

IAS/Park City Mathematics Institute, Institute for Advanced Study

Undergraduate Summer School Participant

Park City, UT

Summer 2023

Learned about quantum algorithms. Also worked on computationally estimating the number of holonomy vectors of at most a certain length on a Veech surface.

Vocational

Department of Mathematics, The University of Chicago

Mentor, Mathematics REU

Chicago IL

Summers 2025, 2026

Advised students in the algebraic geometry and algebraic topology groups

Department of Mathematics, University of Southern California

Grader

Los Angeles, CA

2024–2025

Graded weekly assignments for Calculus III in Fall 2024 and Calculus II (for Engineers and Scientists) in Spring 2025.

Student-Athlete Academic Services, University of Southern California

Undergraduate Tutor, Mathematics and Physics

Los Angeles, CA

2023–2025

Tutored student-athletes at USC in a variety of math and physics classes including the calculus sequence, the intro to physics sequence, number theory, and probability theory.

Community

SC Math Club

Los Angeles, CA

President

2023–2024

Rebuilt the e-board for Spring 2024. Organized events for the undergraduate math community like the department BBQ, the integration bee, and weekly general meetings.

Integration Bee Committee, USC

Los Angeles, CA

Chair

2022–2024

Founded the integration bee at USC. Worked alongside the undergraduate math student association at UCLA to co-organize the first annual UCLA-USC integration bees.

Writing

The stable adjunction in $\mathbb{A}^1$ -homotopy theory

2026

arXiv:2607.14411

Vortex stability in interacting Bose-Einstein condensates

2025

with S. Haas and A. Wirthwein, *Phys. Rev. B* **112**, 104511

Talks

Lecture Series, UChicago Math REU 2026

Chicago, IL

Mathematical Genealogies

July 2026

On critical genealogies in mathematics, with an eye to algebraic geometry

Lecture Series, UChicago Math REU 2025

Chicago, IL

Derived Fun: Towards Homotopical Algebra

August 2025

On derived functors in homotopy theory and the cotangent complex

Volunteer Talk, UChicago Math REU 2024

Chicago, IL

The Where's Waldo of Infinite Loop Spaces

August 2024

Based on recent work of J.P. May, H.J. Kong, F. Zou and discussions with J.P. May

APS March Meeting 2024

Minneapolis, MN

Single Vortex Dynamics in Binary Bose-Einstein Condensates

March 2024

Based on work done with S. Haas and A. Wirthwein

Undergraduate Talk, IAS/Park City Mathematics Institute 2023

Park City, UT

Billiard Dynamics on the Double Pentagon

August 2023

Delivered with H. Malik, S. Rothstein, N. Ringrose, and E. Brodsky. Advised by A. Artiles.

Honors and Scholarships

NSF Graduate Research Fellowship

2025

Awarded by the National Science Foundation

Haltom Sr. Endowed Scholarship and Gleberman Endowed Scholarship

2024

Awarded by USC Dornsife

Lick Scholarship

2023, 2024

Awarded by the USC Dept. of Physics & Astronomy for conference travel

Honorable Mention in the Physical Sciences, Math, and Engineering Category

2023

USC Undergraduate Research Symposium

USC Department of Mathematics Outreach Award

2022

Languages (computer and otherwise)

Computer: Python, Mathematica, C++.

Human: English (native), Tamil (native), French (proficient).

Research Interests

Homotopical algebraic geometry, motivic homotopy theory, algebraic K -theory, moduli spaces, and stable homotopy theory. Also, holography and flux compactifications in string theory.